



Autonomous Infrastructure Threat Intelligence Agent

Bright Data AI Agents Web Data Hackathon

Track 3: Security & Compliance

LangGraph

Claude Opus 4.6

Bright Data

Kafka

github.com/YOUR_USERNAME/sentinel-ai

The Problem

- Kubernetes CVEs drop on the open web
 - no SIEM monitors them
- Credential leaks surface on paste sites
 - days before anyone notices
- Vendor breaches blindside companies
 - no continuous monitoring
- Regulatory changes slip through the cracks
 - manual tracking fails
- A human threat analyst takes DAYS per threat
 - most are never investigated

Two worlds that never connect:
The Open Web $\longleftrightarrow$ **Your Infrastructure**

The Solution

SentinelAI bridges these worlds — autonomously.

1. DISCOVER

Scans the open web for CVEs, advisories, credential leaks, vendor breaches

2. CORRELATE

Matches threats against your actual K8s clusters, Terraform state, AWS resources

3. SCORE

Calculates blast radius, exposure, and exploitability — not just CVSS

4. REMEDIATE

Generates exact kubectl, Terraform, and Helm commands to fix

5. NOTIFY

Sends Slack alerts and creates Jira tickets automatically

"Enter a company name. Watch AI agents investigate in real time."

Zero config. One command. Full threat landscape.

Architecture

Bright Data (Open Web)

SERP API

Web Unlocker

Scraping Browser

Web Scraper API

LangGraph Agent Pipeline

Orchestrator

Discovery

Credential Leak

Vendor Risk

↓ merge results ↓

Correlation

Risk Scoring

Remediation

Notify

Infrastructure

Kafka | PostgreSQL | Docker

Event Bus Checkpoints Containers

Output

Dashboard | Slack | Jira

Real-time SSE Alerts Tickets

Claude Opus 4.6 via LiteLLM

Autonomous reasoning with tool use — agents decide what to scrape and when

Bright Data Integration

All 5 tools used for distinct, critical purposes

SERP API

Searches Google for CVEs, breach news, vendor incidents, credential leaks, regulatory changes

Web Unlocker

Bypasses anti-bot on NVD, AWS bulletins, vendor portals, paste sites, regulatory websites

Scraping Browser

Navigates JS-heavy GitHub Security Advisories, AWS Health Dashboard, SaaS status pages

Web Scraper API

Structured extraction from CVE databases, compliance registries — returns clean JSON

MCP Server

Connects Claude directly to Bright Data — the agent autonomously decides what to scrape

Demo Flow

3 minutes that win

0:00

Enter "Acme Corp"

Agent starts, loads K8s + Terraform inventory

0:30

Discovery agents fan out

3 parallel agents scan web via Bright Data

1:00

First threats appear

"Found 5 CVEs affecting K8s 1.28.2"

1:30

Correlation complete

"2 of 3 clusters run affected versions"

2:00

Risk scored

"85/100 — internet-facing, no network policies"

2:30

Remediation generated

"kubectl upgrade + Terraform patch ready"

3:00

Auto-notify

Slack alert + Jira ticket created live

The audience sees the full pipeline in 3 minutes.

Pre-cached demo data ensures instant, reliable results.

Tech Stack

- **Docker + Docker Compose**
One command to launch the entire stack
- **LangGraph**
Stateful graph-based multi-agent orchestration
- **Claude Opus 4.6 via LiteLLM**
Most capable reasoning model, unified API
- **FastAPI (Python)**
Async backend, SSE streaming
- **Next.js 14 + Tailwind**
Real-time dashboard with live agent trace
- **Kafka (Confluent 7.6)**
Event-driven pipeline, auditable event bus
- **PostgreSQL 16**
Threat history, LangGraph checkpoints
- **Bright Data SDK (all 5 tools)**
Full open-web access with anti-bot bypass
- **Slack SDK + Jira REST API**
Automated alerting and ticket creation

Why SentinelAI Wins

Novelty

Not a generic LLM wrapper — infrastructure-aware AI that correlates web threats against YOUR environment

Domain Expertise

Built by someone who runs K8s, Terraform, and AWS in production — real SRE workflows

Bright Data Integration

All 5 tools used for distinct, critical web intelligence tasks

Production Architecture

Kafka event bus, Docker, environment parsers — this could ship today

Demo Impact

CVE drops → agent finds it → checks clusters → generates fix → alerts Slack — in 3 minutes

Real-World Value

\$15B+ threat intelligence market. Existing tools (Snyk, Wiz) cost \$50K+/year



Autonomous Infrastructure Threat Intelligence

Try it: `docker-compose up -d`

Open <http://localhost:3000> and investigate.

Built with

Bright Data

LangGraph

Claude Opus 4.6

Kafka

Docker

Thank you!